Panlin Li (Direct Ph.D. Application)

High-Dimensional Statistical Learning | Data Modeling

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Member, China Society for Industrial and Applied Mathematics

Education

Xinjiang Agricultural University, School of Mathematics & Physics — Sep. 2023 – Jul. 2027

B.S. in Mathematics & Applied Mathematics · GPA 3.86 / 5.0 · Top 2.5% of major · Advisor: Prof. Hua Huang

- Honors: Outstanding Volunteer (×9) & Outstanding Team, CYL Central Committee; Hematopoietic Stem Cell Donation Honor Certificate; Outstanding Student, Innovation & Entrepreneurship Star (2 consecutive years), Science & Innovation Pioneer Scholarship — XJAU.
- Key outcomes: published 12 papers (9 first-author), with 1 more first-author SCI Q1 manuscript under review; led or participated in 15 student innovation projects (2 national, 6 provincial), one recognized by Huawei Research Institute; 4 utility-model patents and 4 software copyrights as first inventor; 5 national & 7 provincial competition awards; volunteer work covered by *China Daily*, *People's Daily*, and other media outlets.

Research & Projects

Xinjiang Natural Science Foundation · General Program · Participant — Functional Statistical Modeling via Spatial-Spectral Feature Fusion of Hyperspectral Imaging — 2026 – Present

- Working on the project's two key scientific questions — integrated spatial–spectral representation, and functional statistical modeling of the fused signal. Curated multi-cultivar HSI/NIR datasets with spectral correction, outlier removal and common-wavelength grid alignment; built multi-basis functional representations, applied regularized modeling in a high-dimensional function space, and paired it with computable identifiability diagnostics so the model's validity range is testable rather than merely asserted.

National Innovation Project · Ongoing · Participant — Apple Origin Traceability via Machine Learning & Transfer Learning — 2025 – Present

- Lead-authored the proposal; led cross-year apple HSI and image data collection across Xinjiang / Shandong / Gansu; built a 9-domain benchmark with an origin × year leakage-controlled evaluation protocol and ran PLS-DA / SVM / 1D-CNN / Transformer baselines; built a benchmark of 1,950 fruits, 4,529 spectra and 229 bands (591–1,001 nm) spanning 3 harvest years and 9 acquisition domains; proposed a leakage-free protocol (leave-one-harvest-year-out with a fold-restricted encoder) that measures and corrects the cross-year inflation caused by a shared encoder.

National Innovation Project · Completed · PI — Small-Sample Time-Series Forecasting for Energy Economics — 2024 – 2025

- Addressed limited accuracy under data scarcity and short series; benchmarked meta-learning, data augmentation and transfer learning, then built a Transformer-plus-few-shot forecaster and validated it against controls.

Other projects: Wenmai Zhiyuan Hub (Provincial · Excellent · PI) · Digital Cotton Field (Provincial · Ongoing · PI) · Multimodal Navigation for the Visually Impaired (Provincial · Ongoing) · Apple SSC Prediction via FDA + Image Fusion (Provincial · Excellent) · Nonlinear Wave Solutions of High-Dimensional PDEs (University · Excellent · PI), among others.

Publications

- Li P.L., Li Y.C., Li L.J., et al. *The label economics of calibration transfer: a five-benchmark, dual-modality evaluation of simple corrections versus deep and physics-informed representations*. Chemometrics and Intelligent Laboratory Systems. (SCI Q1; IF: 3.8; Under review)
- Li P.L. (presenter), Yang S.Z., Huang H.* *Upper bound of reference-value reliability and a sampling design table for near-infrared calibration of apple SSC*. The 11th National Conference on Near-Infrared Spectroscopy, China.
- Li P.L., Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.
- Li P.L., Bao F.C., Zhang J.Y., et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- Li P.L., Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- Li P.L., Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- Li P.L., Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- Li P.L., Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004.
- Li P.L., Li J., Kadir A., et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Rao W.J., Li P.L. *An improved kernel regularized nonhomogeneous grey model based on conformable fractional accumulation*. Transactions on Computational and Applied Mathematics, 2024, 4: 137–147. (Co-First Author)
- Zhang N., Li P.L., Bao F.C., Liu X.Y. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.

- Jiao Q.T., Li P.L., Mei T., et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- Huang H., Wang M., Liu Y., Li P.L., et al. *Prediction of Apple Soluble Solids Content Using a Functional Partial Linear Regression Model Fusing Image Color Features and Vis-NIR Spectra*. Spectroscopy and Spectral Analysis, 2025, 45(S1): 415–423.
- **Utility-Model Patents (4, as first inventor):** Section-Connection Component for a White Cane · Portable Ultrasonic Detection Device · Pesticide Spraying Drone · Foldable Anti-Collision Spraying Boom for Drones.
- **Software Copyrights (4, as first author):** Intelligent White Cane Vision Recognition System · Smart Cane Data Collection & Monitoring System · User Behavior Analysis and Recommendation System · Big-Data-Driven Market Analysis and Forecasting System.

Competitions & Awards

- China International College Students' Innovation Competition (2024) · Bronze Award *National* 2024.10
- 11th National Undergraduate Statistical Modeling Contest · Third Prize *National* 2025.10
- "Chia Tai Cup" National Market Survey & Analysis Contest (15th & 16th) · Third Prize × 2 *National* 2025 – 2026
- DataBee Cup National Interview Survey Contest · Third Prize *National* 2024.10
- 11th National Undergraduate Statistical Modeling Contest, Xinjiang Division · First Prize *Provincial* 2025.08
- "Chia Tai Cup" Market Survey & Analysis Contest, Xinjiang Division · First Prize *Provincial* 2025.04
- National Mathematical Modeling Contest, Xinjiang Division · Second Prize *Provincial* 2024.11
- Plus 4 further provincial awards (China International College Students' Innovation Competition 2025 Bronze; 14th China Innovation & Entrepreneurship Competition Third Prize; 10th Statistical Modeling Contest Second Prize; Information Literacy Competition Excellence Award).

Student Experience

Class Monitor, Mathematics 2302, School of Mathematics & Physics 2023.09 – 2027.07

Class Monitor, 19th Youth Marxist Training Program, Xinjiang Agricultural University 2025.05 – 2026.05

Head, Volunteer Association, School of Mathematics & Physics, XJAU 2024.09 – 2025.11

- Founded the 20-member "Shield of the Future" volunteer team; took part in 11 public-welfare programs organized by the CYL Central Committee and the China Young Volunteers Association across Shandong, Henan, and Xinjiang; led the class to the university's *Outstanding Class Collective* honor.

Skills & Self-Assessment

- **Language:** TOEFL 110/120; first-authored multiple English-language papers; able to write and submit academic English independently.
- **Professional:** high-dimensional statistical inference (debiased Lasso, double machine learning, generalized random forests), functional data analysis (FPCA, function-on-function regression), tensor decomposition & multimodal fusion; Python (PyTorch / scikit-learn), end-to-end modeling from theory to implementation to validation. Tools: LaTeX · Git
- **Self-assessment:** Photographer and outdoors lover with a strong exploratory drive, hands-on ability, and execution. Through 15 projects and 12 papers I have walked the full loop from mathematics → statistics → machine learning → real-world deployment — which made clear that my bottleneck is theoretical depth. In the Ph.D. stage I aim to move from application-driven work toward more theoretical methodology, while staying grounded in real applications.